

Pneumonia detection using VGG16 model

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Abstract - Pneumonia is a lung infection which identifies the inflammation of the air sacs in the lung, which causes fluid to build up in the lungs and makes difficulty in breathing. It is one of the most common and serious health problems in the world, especially in young children, older individuals and people who are more open to infectious diseases due to their weakened immunity systems. Early detection of pneumonia is essential for effective treatment, the manual interpretation of Chest X-rays is often slow and requires trained medical personnel that may not be available in all settings. In this study, a deep learning-based pneumonia detection system is built. The model is built using transfer learning with the VGG16 architecture, where the pre-trained convolutional layers are used as feature extractors, and a custom classification head is trained for identifying normal and pneumonia-affected chest X-ray images. The kaggle chest X-Ray Pneumonia dataset is used, and images are subjected to resizing, normalization, and data augmentation that enhances model generalization. First, training starts with frozen base layers and subsequently fine-tunes the upper layers. As a result, an accuracy of 89.74% is demonstrated on the test set. Finally, visualization techniques will be suggested to further identify regions in the X-ray influencing the model's prediction. Results point out that transfer learning can serve as a good approach in pneumonia detection and, with explainable outputs, may provide clinical decision support.

Keywords - Chest X-ray; Convolutional Neural Network; Transfer Learning; Pneumonia; Model Interpretability; Classification; Deep Learning

I. INTRODUCTION

Pneumonia is an infection of lungs where the air sacs (or alveoli) of the lungs fill with fluid (or pus), which makes breathing difficult. This can be caused by viruses, bacteria, or fungi and is most dangerous for children under five, and for people with weak immune systems, and for older adults. Despite improvements in health services, pneumonia remains a top public health concern. According to reports from UNICEF and WHO in recent years, more than 700,000 deaths among children of age five and under annually are attributed to pneumonia-that is, it remains the most infectious cause of children's mortality in the world. Globally, it is estimated that over 150 million new cases of

pneumonia occur every year. The disease does not only bring high mortality but also a long hospital stays, long-term respiratory weakness, and huge health burdens in resource-poor settings. The key for its effective treatment is the early diagnosis, but traditional manual reading of chest X-rays may be time taking and require expert radiologists, especially in rural areas or overcrowded medical facilities.

- Developing a deep learning-based to detect pneumonia model using VGG16 architecture through transfer learning.
- To preprocess chest X-ray images through resizing, normalisation and data augmentation so that the model generalises better.
- To train the model in two stages, beginning with frozen convolutional layers and later fine-tuning higher-level layers at a lower learning rate for improved accuracy.
- To evaluate the model using precision, accuracy, F1-score, recall, and a confusion matrix on a test set.
- To enhance model interpretability by identifying key regions influencing predictions.

The organization of the remaining sections of this paper is as follows: Section 2 presents the related literature on pneumonia detection and deep learning methods. Section 3 describes the dataset, preprocessing steps, model architecture, and training strategy. Section 4 presents the results, including evaluation of metrics and the confusion matrix. Section 5 discusses the future scope and potential improvements of the model.

II. RELATED WORK

Many researchers have applied deep-learning models to detect pneumonia from chest radiographs. Rajpurkar and colleagues developed CheXNet, a model built upon the DenseNet architecture, which achieved diagnostic performance comparable to expert radiologists on large-scale datasets. Kermany et al. demonstrated the usefulness of the transfer learning with pre-trained convolutional neural

networks when applied on pediatric X-rays. Stephen and co-authors improved accuracy of VGG16 model through systematic data augmentation, while Sharma's work with EfficientNet reported improved sensitivity for the identification of infected cases. Together, these works reflect the increased performance of convolutional models in automating radiographic diagnosis. Dutta et al. explored several ResNet architectures and emphasized the role of deeper layers in improving feature extraction. Patel et al. used MobileNet to facilitate lightweight deployment in resource-poor healthcare settings. Thomas et al. compared several CNN models and concluded that transfer learning outperforms models trained from scratch by a wide margin. Mahesh et al. focused on binary classification based on AlexNet but report limitations owing to dataset imbalance. Naveen et al. explored SVM + CNN features and reported moderate accuracy without model generalization. Several other works explored pre-trained architectures including VGG19, InceptionV3, and Xception for pneumonia detection-all reporting promising accuracy but mostly limited to small datasets with a pediatric focus.

Even though the above mentioned studies have shown very strong results, but still, it has several gaps remain in the them like:

- Most of the works have used small and imbalanced datasets, often pediatric-only, which limits generalization to real-world populations.
- Most of the models focused only on the classification accuracy, without giving visual explanations of why such a decision was made.
- Explainable AI techniques were rarely used, making it difficult for doctors to trust and adopt these models in clinical settings.
- Evaluation of metrics beyond accuracy, such as ROC-AUC, specificity, and calibration, were not consistently reported, reducing reliability for medical decision support.

The present work aims to address these challenges by applying transfer learning with the VGG16 architecture on a balanced, augmented chest X-ray dataset.

The interpretability and visualization techniques are incorporated to make sure to highlight the regions of the lung which influences the model's predictions, which in turn improves lucidity and usability. And at the end, the final model gives good performance and reliability when tested on a test set and validation set. Integration of explainability and structured evaluation will render the system more practical and interpretable clinically in real-world use.

III. METHODOLOGY

A. Dataset Description

For model training and testing, the publicly available "Chest X-Ray Pneumonia" dataset created by Paul Mooney on Kaggle was used. The dataset contains 5,680 pediatric chest-X-ray images, and each image is labelled as either normal or

pneumonia. The images are already stored in three different folders for training, validation and testing. There are 4,172 images in the training set, 1,044 images in the validation set and 624 images in the test set, with more pneumonia cases than normal cases in each split. Because of this class imbalance, it is important to apply data-augmentation techniques during training.

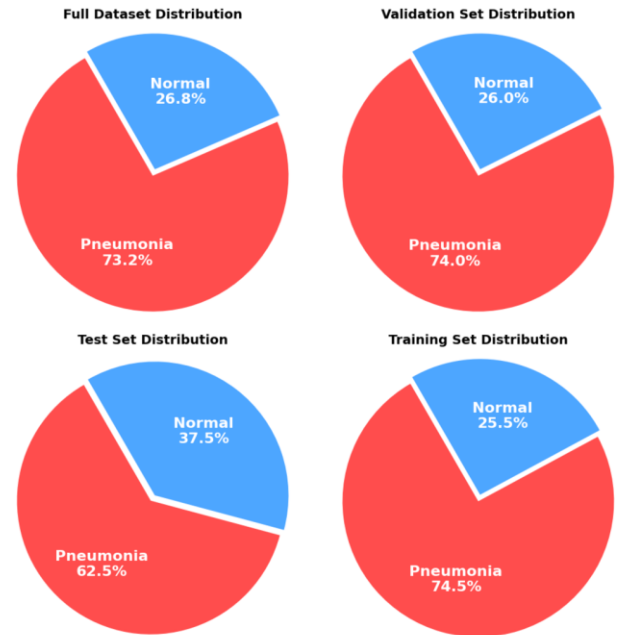


Figure 1. Dataset class distribution across validation, training and test sets.

This dataset contains an imbalance in the dataset, with a bigger number of differences between pneumonia and normal cases. The training dataset includes 3110 Pneumonia and 1062 Normal images, while the validation dataset includes 773 Pneumonia and 267 Normal images and the test dataset includes 390 Pneumonia and 234 Normal images.

All X-ray images were pre-processed to a standard size and intensity range before modeling. All the images were changed to standard 224 x 224 pixels and converted to 3 channel RGB format (even though the original scans are grayscale). Then scaled pixel intensities to [0,1] range by dividing 255, normalization that ensures consistent input scale for the neural network.

In the given pneumonia dataset, there is an imbalance training data, therefore the data ,augmentation was applied using keras ImageDataGenrator on training the X-ray images only. The below mentioned augmentations were applied during training:

- Rotation: up to ± 25 degree around the image center
- Horizontal shift: up to $\pm 15\%$ of the width
- Vertical shift: up to $\pm 15\%$ of the height
- Zoom: up to $\pm 15\%$
- Shear: up to ± 10 degree
- Horizontal flip: applied with 50% probability

Rescaling and resizing were applied only to the training dataset, to introduce variability without altering the label and to help the model generalize even better in finding the results.

B. Proposed Architecture

Transfer learning is used in the suggested setup; the VGG16's pre-made structure pulls features from a deep network that was trained on ImageNet's thousand classes. While that version sorts of broad image types, here the focus is on spotting pneumonia versus normal scans, just two outcomes. So instead of keeping everything intact, adjustments are made to reshape how VGG16 handles the patterns, shifting its strength to analyzing X-rays of lungs more effectively.

i. Features Extraction (VGG16 Convolutional Base)

The conv part of the VGG16, which consists of 13 conv layers with 5 max pooling stages, is retained with its original ImageNet tuning.

Instead of beginning from scratch, it employs all the previously learned levels to detect simple visuals such as lines, surface details and shapes, which is useful in identifying abnormalities of the lungs or other body structures on x-rays. At the outset of the fine-tuning process, the full set of convolutional blocks are frozen so that the core detection capabilities remain stable and there are no changes early that will create confusion.

ii. Replacement of Fully Connected Layers

The original VGG16 fully connected (dense layers) classifier is removed and replaced with an architecture that is more compact and efficient for binary medical classification. The original architecture takes feature maps and flattens them processing into a dense layer. This architecture replaces this step with a Global Average Pooling (GAP) layer that reduces spatial dimensions to a single vector by averaging feature responses, which reduces the total number of parameters and minimizes the risk of overfitting.

The GAP layer is followed by a Batch Normalization layer that improves gradient flow and stabilize training to ensure that the custom classifier learns effectively from the extracted features.

iii. Custom Classification Head

A new classifier head is constructed with multiple fully connected layers:

- A Dense layer with 256 units and ReLU activation
- Dropout (50%) to eliminate co-adaption and overfitting
- A Dense layer with 64 units and ReLU activation
- Dropout (30%) for additional regularization

The final layer is a 2-unit dense layer that corresponds to two classes (Pneumonia and Normal), which is followed by Softmax activation that converts the raw score into class probabilities.

iv. Fine-Tuning Strategy

After training the head of the custom classifier, selective fine tuning is performed. The top 10 convolutional layers of VGG16 are unfrozen and allowed to update their weights.

This step helps the model to generalize better chest X-Ray patterns such as infiltrates, opacities or consolidations.

Fine-tuning is performed with lower learning rate (1×10^{-5}) to avoid changes in the pretrained weights significantly but allow slight changes to improve the accuracy when fine-tuning for a new domain.

v. Training Configuration

Adam optimizer and Categorical Cross-Entropy loss function were used to train the model. The model's initial learning rate started at 1×10^{-4} and was automatically reduced during the training using the ReduceLROnPlateau callback when the validation performance plateaus.

vi. Compared with the original VGG16

The updated design is not like the old V-G-G-16; it ditches bulky fully connected parts, uses global average pooling instead, throws in batch norm along with dropout to prevent overfitting, while only tweaking certain layers. Because of these changes, learnable weights drop sharply from about 138 million down to just 15 or 20 million, which means less processing power needed and better performance on medical scans.

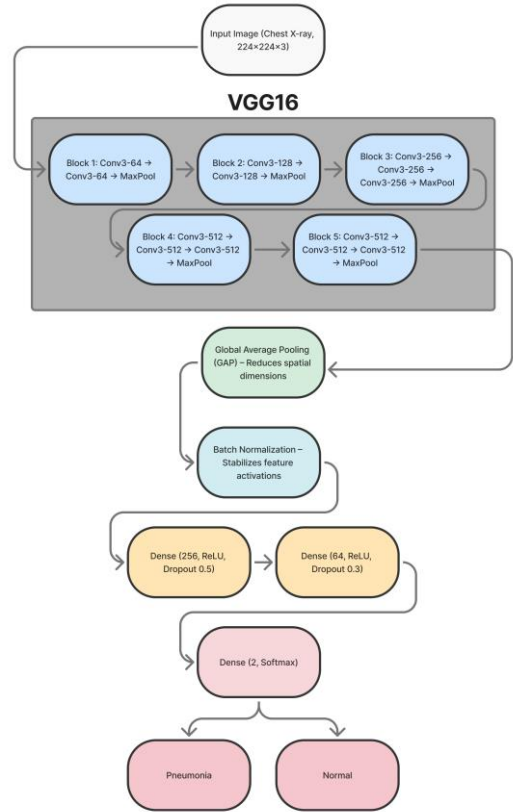


Figure 2. Modified VGG16 architecture for pneumonia detection using transfer learning.

IV. RESULTS & DISCUSSION

This section shows the results of the achieved setup and performance of the modified version of the VGG16-based pneumonia detection model.

A. Model Performance

The performance of the model was first evaluated on the training dataset, and then on validation and test datasets using accuracy, precision, recall, and F1-score as metrics. The fine-tuned VGG16 model achieved 96.19% accuracy on the training dataset, 95.79% accuracy on the validation dataset and 89.74% accuracy over the test dataset. The final results indicate that the model is generalized well and effectively classifies normal patient's X ray and Pneumonia patient's X ray.

The confusion matrix highlights that most of the pneumonia cases were identified correctly, with a few misclassifications between the two categories. The model demonstrates a stronger sensitivity (recall) toward pneumonia detection, which is desirable in a medical diagnostic context to minimize false negatives.

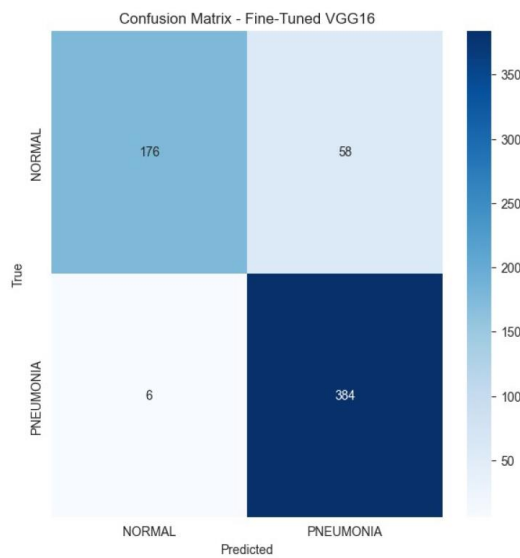


Figure 3. Visual representation in the form of heatmap confusion matrix of the results of the test dataset.

B. Training and Validation Analysis

During our two levels of training process, the accuracy and loss trends are represented using the learning curves in figure 5.

Accuracy gets better as the number of epochs increases. With appropriate learning rate and regularization strategies, the model shows very less overfitting and slowly converges.

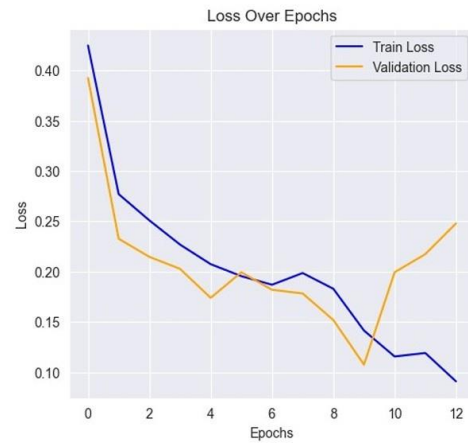


Figure 4. Loss curve of training and validation loss

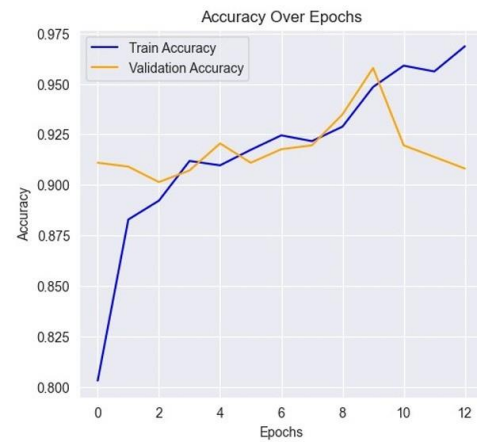


Figure 5. Accuracy curve of training accuracy and validation accuracy as number of epochs increases

COMPARATIVE ANALYSIS

Study	Architecture	Accuracy (%)	Remarks
[1]	Vision Transformer-based model	90.3	Higher accuracy but needs larger datasets and heavy computational cost
[2]	Deep CNN transfer learning	87.6	It has good Performance but it lacks visual representation.
[4]	Xception-based CNN+ Logistic Regression	88.1	It has good sensitivity and has higher complexity
[5]	Comprehensive CNN survey models	87.0	It summarizes existing CNNs but has no experimental evaluation.
[10]	Lightweight CNN	85.7	It has Faster Interface with reduced complexity, but lower diagnostic accuracy compared to others

	Proposed Model(VGG16 + Transfer Learning)	89.74	It has balanced accuracy and lower parameter count, also it has two stages of fine-tuning, and has built-in interpretability
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Table I. Comparison table of our model with different models

Comparative results from [1, 2, 4, 5, 10] are presented, showing that most models achieve 85-90% accuracy, often based on large datasets or with heavy architectures.

Vision Transformer in [1] achieved the highest accuracy, which requires substantial computational resources and large pre-training data.

The deep CNN model of [2] and the Xception-based approach in [4] are the strongest performers in terms of classification, but suffer from poor interpretability, seriously limiting clinical transparency.

The review of CNNs in [5] also finds that many perform well but that most research simply shows raw accuracy without considering explainability.

The lightweight CNN approach in [10] allows for faster inferences and fewer parameters but at the cost of reduced diagnostic precision.

The proposed VGG16-based transfer learning model achieves 89.74 % accuracy, comparable or higher than the works that were cited in this paper, and keeps the computational footprint modest.

The two-stage training stabilizes convergence and reduces overfitting of the proposed model .

Thus, this combination of efficiency, interpretability, and reliable accuracy presents the proposed model as one balanced alternative to heavier architectures reported in [1], [2], [4], [5], and [10].

GitHub repo link :

<https://github.com/SujayVeershette/Pneumonia-detection-using-VGG16-model-.git>

V. CONCLUSION & FUTURE SCOPE

This work demonstrated that a transfer-learning approach on the VGG16 backbone can help in classifying chest X-ray images reliably into classes such as pneumonia and normal. Pre-processing, class-imbalance handling, and the strategy of two-stage training enhanced stability, reducing overfitting. Architecture modification by replacing the original fully connected layers with global average pooling, batch normalization, and dropout reduced the number of parameters while maintaining the powerful representation ability of the network. The best model achieved 89.74% accuracy on the test set with high recall for the class pneumonia, at the cost of a lower precision, which reduces the risk of false-negative clinical outcomes. A comparative analysis confirmed the competitiveness of the proposed method while retaining lower computational cost compared to heavier architectures like Vision Transformers or deeper CNNs. Clinical relevance was further strengthened by including interpretability methods that highlighted the

regions of the images that are influencing the predictions. Overall, our results demonstrated that transfer learning using VGG16 remains practical, resource-efficient, and clinically aligned for automated pneumonia detection from chest radiographs.

FUTURE SCOPE

This present work demonstrates the effectiveness of VGG16 transfer learning to discover pneumonia in patients using chest X ray images. However, there is a lot of directions which are still open for further improvement and investigation:

1. The Addition of more Advanced Deep Learning Architectures

Future work may explore more recent and superior architectures like EfficientNet, DenseNet, MobileNetV3, or even Vision Transformers. These more modern architectures outperform the older backbone VGG16 on accuracy, inference speed, and computational complexity.

2. Training on Bigger and Multiple Datasets

The dataset can be expanded to include images of X-ray from multiple hospitals, cities, countries, and imaging devices to reduce domain bias and to slowly increase the capability of the generalization of the model. With bigger and more varied datasets would eventually make the model learn more robust patterns applicable in real clinical settings.

3. Explainable AI Integration -

XAI-With the inclusion of interpretability methods such as LIME or layer-wise relevance propagation, one can visualize exactly which regions of the lungs are contributing to specific predictions. Such transparency is required for clinical trust and will allow radiologists to validate or cross-check the decision-making process of the model.

4. Addressing Class Imbalance and Rare Pneumonia

Future work may explore focal loss, class-balanced training, generating synthetic images with GANs or diffusion models, and anomaly detection approaches in order to better deal with rare cases of pneumonia. Even if there is an uncommon disease pattern, then the model can generalize it in a better reliable way with more confidence with these methods.

5. Longitudinal Monitoring of Patients

When the future model is upgraded to read multiple X-rays which are tracked for a long time, the model can also track how the disease has changed over time. This helps doctors detect problems in the earlier stage of the disease and see if their treatment is doing any effects on the patient

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